

A large **tropical cyclone** releases as much **energy** as **half of Earth's electrical generating capacity**

Cyclones

The intensity and scale of cyclones makes them the most destructive phenomena in Earth's atmosphere

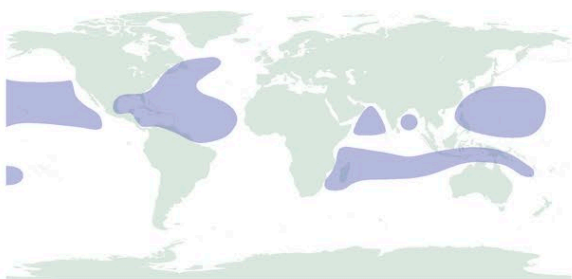
A cyclone is a rotating storm system centered around an area of low pressure. It forms as warm, moist air rises, causing a fall in pressure, which draws in more moist air from all directions. An effect known as the Coriolis force causes the air to spiral, counterclockwise in the Northern Hemisphere and clockwise in the Southern Hemisphere.

Tropical cyclones

In temperate regions, cyclones are often called depressions and are only rarely destructive. They are at their most intense in the tropics. They are called hurricanes in the North Atlantic, cyclones in the Indian Ocean, and typhoons in the western Pacific. A tropical cyclone forms over the ocean in the presence of slack winds and surface temperatures of more than 79°F (26°C)—usually in late summer or early fall. It becomes a full cyclone when winds exceed 73 mph (118 kph), although sustained speeds can exceed 200 mph (320 kph). Combined with torrential rainfall, this can cause flooding and mudslides and bring down buildings. During their most vigorous phase, cyclones track west. They begin to die only once they make landfall and their supply of warm ocean water is cut off.

TROPICAL CYCLONE REGIONS

Most tropical cyclones develop in two bands between 10 degrees and 30 degrees north and south of the equator. At higher latitudes, ocean surface temperatures are unlikely to be warm enough to fuel the storm's development. Closer to the equator, the Coriolis force will not be great enough to set circulation in motion.

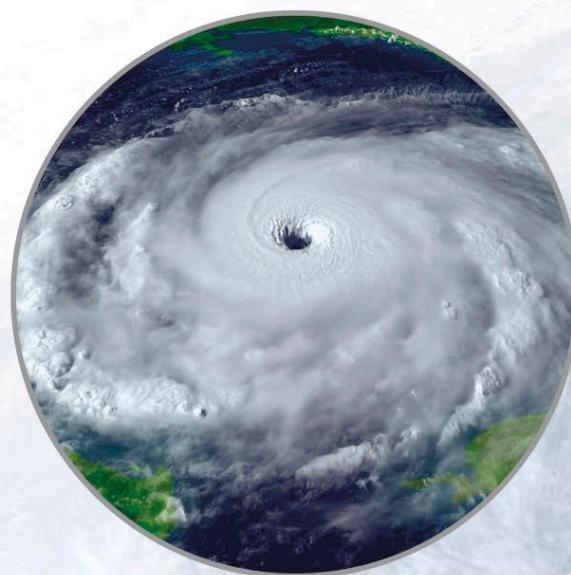


▼ TROPICAL CYCLONE

A section through a tropical cyclone reveals its central eye, where there is an area of sinking air, surrounded by a series of cloudbands.

▷ MATURE STAGE

Seen from overhead, a full-blown tropical cyclone, such as Hurricane Rita, will have a roughly circular cloud pattern, as well as a distinct eye and spiral rainbands.



Flat top

The top of the storm may be as high as 9 miles (15 km) above sea level.

Downpours

The area beneath a cloudband is subjected to heavy rainfall, strong winds, and lightning.



◁ FORCE OF NATURE

Palm trees bend in intense winds as Hurricane Dennis hits the coast of Florida in July 2005, although the storm surge created more damage than the winds.

Cloudbands

Spiralling around the eye are cloudbands that can extend for hundreds of kilometers from the cyclone center, gradually diminishing in height.